AnnotateBench: How Much Labeled Data Do Text-Classification Annotation Strategies Need?

Anote AI

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Abstract

Annotation budgets are a practical bottleneck in applied natural language processing, yet teams often choose labeling strategies without direct evidence about expected accuracy gains, diminishing returns, or cost tradeoffs. We introduce AnnotateBench, a benchmark framework for comparing text-classification annotation strategies through downstream learning curves, cost-performance frontiers, and budget recommendations. This study focuses on public gold-labeled text-classification datasets rather than newly collected human annotations. In the gold-label benchmark, annotation is simulated by revealing labels from the training split according to each selection strategy. The implementation targets a ten-dataset classification suite spanning financial sentiment, question classification, banking intent, news/topic classification, review sentiment, social-media sentiment, and emotion classification. We compare random sampling, uncertainty-based active learning, diversity-based active learning, and a hybrid strategy using a common TF-IDF logistic-regression classifier, and add a seed-0/1/2 API-backed LLM annotator extension that labels selected training examples before fitting the same downstream model. AnnotateBench provides a reproducible empirical framework for measuring how many labeled examples are needed to reach strong downstream performance under different annotation strategies.

1 Introduction

Labeled data remains one of the most expensive inputs to real-world NLP systems. Even when model architectures and training code are standardized, teams still need to decide which examples to annotate, how large an initial labeling batch should be, and when further annotation is no longer worth the marginal cost. These decisions are often made using rules of thumb or isolated active-learning experiments rather than a benchmark that directly compares annotation strategies under a shared budget and evaluation protocol.

AnnotateBench addresses the following question: how do different annotation strategies perform across NLP tasks and label budgets, and how many labeled examples are needed to reach strong downstream performance? The benchmark is designed around learning curves rather than single-budget scores. For each strategy, it evaluates downstream performance as a function of labeled examples, estimates annotation cost, and identifies Pareto-efficient budget-strategy choices.

The paper is motivated by prior Anote work showing that labeling a small number of high-value examples can improve classification performance across datasets such as Financial PhraseBank, Banking, Craigslist, TREC, and Amazon Reviews [18]. AnnotateBench extends that idea into a systematic benchmark across annotation strategies, datasets, label budgets, and cost-performance tradeoffs. The benchmark uses public datasets with existing gold labels and simulates annotation budgets by subsampling labeled training examples selected by each strategy. This design avoids new human-subject data collection while preserving the core experimental question: which examples would an annotation process choose under a fixed budget, and what downstream model quality would those labels support?

This JDSE 2026 submission presents a text-classification benchmark. Its measured scope is public text-classification datasets, five label budgets, four gold-label selection strategies, a TF-IDF logistic-regression downstream model, and publicly calibrated human-annotation cost assumptions. The implementation also includes a seed-0/1/2 LLM annotator extension in which selected training examples are labeled by an OpenAI model, then used to train the same downstream classifier. LLM confidence, ECE, and LARI are used as diagnostics for this extension and for row-level reliability case studies rather than as the central contribution. The paper does not claim measured results for NER, QA, summarization, or relation extraction, which require task-specific data structures, annotation units, and metrics. Future work will extend this grid to stronger models, measured annotation costs, and broader LLM-assisted annotation baselines.

2 Related Work

2.1 Data-Centric AI Benchmarks

Recent benchmark work argues that progress in applied AI depends not only on model architecture, but also on how data is acquired, filtered, labeled, and evaluated. DataPerf formalizes data-centric AI evaluation through benchmark tasks for data selection, debugging, acquisition, and related dataset-improvement workflows [10]. Data acquisition benchmarks further emphasize transparent pricing, standardized data formats, and realistic interactions between data providers and consumers [2]. DataComp makes a related point at multimodal scale: dataset filtering and curation can be benchmarked under standardized training and evaluation pipelines [3]. AnnotateBench follows this data-centric benchmark tradition, but focuses specifically on label-budget allocation for NLP annotation strategies.

Data valuation work is also closely related because it asks which individual examples are useful for model training. Dataset Cartography uses training dynamics to identify easy, ambiguous, and hard-to-learn examples in NLP datasets [14]. OpenDataVal provides a unified benchmark for data valuation algorithms across image, natural-language, and tabular tasks [5]. Transferred Shapley approaches adapt data valuation to fine-tuning large language models [13]. AnnotateBench differs by evaluating examples at the moment they would be selected for annotation, before their labels are assumed to be available.

2.2 LLM-Assisted Annotation

Large language models have made automated and semi-automated annotation a live empirical question rather than a speculative future direction. Wang et al. show that GPT-3-generated labels can substantially reduce labeling costs for downstream NLP tasks, especially when combined with limited human labels [19]. A recent survey organizes the area around LLM-generated annotations, assessment of those annotations, and downstream use of LLM-labeled data [15]. Empirical studies report that LLM labels can be competitive with human labels for some text-annotation tasks [4], that LLM-generated training labels can support supervised text classifiers across computational social science tasks [11], and that teacher-student pipelines can use GPT-labeled data to train smaller multilingual news-topic classifiers [7]. These results motivate including LLM annotation in the full AnnotateBench grid, but they also show why LLM baselines must be versioned, costed, and evaluated under the same budget protocol as human annotation strategies.

Most LLM-as-annotator studies ask whether model-generated labels can match or replace human annotations. They commonly report label accuracy, human–LLM agreement, or downstream classifier performance, and some include direct annotation-cost estimates. Fewer studies jointly vary the label budget, the example-selection strategy, and the annotation source, then evaluate the trained downstream model under a shared cost model. Calibration and confidence-aware review are also usually treated as secondary diagnostics rather than as part of the central budget-allocation problem. This leaves a practical gap for data-science teams: when the labeling budget is fixed, should examples be labeled by humans, LLMs, active selection, or a hybrid strategy, and how many human labels are actually needed before additional annotation has diminishing returns?

2.3 Few-Shot Classification with Human Feedback

AnnotateBench is closest in spirit to recent work on improving classification with a small number of labeled examples. Vidra et al. evaluate a continuous human-feedback loop across Financial PhraseBank, Banking, Craigslist, TREC, and Amazon Reviews, using models such as GPT-3.5, BERT, and SetFit [18]. SetFit is especially relevant because it demonstrates strong few-shot text classification without prompt engineering or very large models [17]. AnnotateBench builds on this line of work by turning the feedback loop into a benchmark question: at each budget, which strategy should choose the examples, what does that strategy cost, and where does it sit on the learning curve?

2.4 Human-in-the-Loop and Active Annotation

The most relevant recent work is no longer only classic uncertainty sampling; it is the combination of active selection, LLM-generated labels, and selective human review. Rouzegar and Makrehchi integrate LLM outputs with human annotation inside an active-learning framework and explicitly optimize cost-performance tradeoffs for text classification [12]. This line of work is close to AnnotateBench's motivation, but it is primarily method-specific. AnnotateBench is positioned as a benchmark layer: it compares multiple annotation strategies, budget levels, and cost frontiers under one reproducible protocol.

2.5 Expert, Span, and Domain-Specific Annotation

Newer LLM annotation work also cautions against treating LLM labeling as uniformly solved. Span-annotation studies find that LLMs can be cost-efficient and sometimes comparable to skilled annotators, but agreement varies by task and model type [6]. Expert-domain evaluations report mixed results in finance, biomedicine, and law, including limited gains from reasoning-style inference-time techniques in some settings [16]. AnnotateBench therefore treats LLM annotation as a strategy family whose cost, reliability, and domain sensitivity must be benchmarked rather than assumed.

2.6 Benchmark Datasets

The classification suite includes Financial PhraseBank, TREC question classification [8], and Banking77 intent classification [1], plus additional public text-classification benchmarks for news/topic classification, long-form topic classification, review sentiment, social-media sentiment, and emotion classification. This selection is intended to stress the same budgeted classification pipeline across different label-space sizes and domains, while avoiding claims about structured prediction or generation tasks that are not yet measured.

Table 1: Representative evidence matrix for related work. Prior studies motivate LLM-assisted annotation, active selection, and data-centric benchmarking, but they rarely combine budget sweeps, strategy comparison, downstream model evaluation, cost frontiers, and disagreement handling in one protocol.

Work or area	Main evidence	Budget / strategy?	Cost or disagreement?	Gap addressed by AnnotateBench
Data-centric benchmarks [2, 3, 10]	Benchmark scores, standardized data workflows, and acquisition settings	Partial	Cost-aware in some settings, but not annotation Pareto frontiers	Narrows data-centric benchmarking to NLP annotation strategy selection under explicit label budgets.
Data valuation [5, 13, 14]	Example value and retraining performance	Selection focused	Usually limited	Measures value at annotation time, before labels are assumed to be available.
Few-shot and human-feedback classification [17, 18]	Classification accuracy or macro F1 under small labeled sets	Label-count sweeps	Limited annotation-cost modeling	Extends few-shot feedback results into strategy comparisons, learning curves, and cost frontiers.
LLM annotation studies [4, 7, 11, 15, 19]	Label quality, agreement, downstream accuracy, and surveys of risks	Limited budget variation	Some cost estimates, rarely Pareto	Evaluates LLM labels as one budgeted strategy rather than the only intervention.
LLM + human active annotation [12]	Accuracy, F1, and method-specific cost-performance	Yes	Includes human review	Generalizes a close method-specific setup into a reusable benchmark grid.
Expert and span-level LLM annotation [6, 16]	Agreement, accuracy, cost, and domain quality	Usually task/model focused	Agreement variation is central	Treats reliability and domain sensitivity as benchmark dimensions, not assumptions.
Benchmark classification datasets [1, 8, 9]	Public task-level model scores	No annotation strategy study	Public gold labels	Reuses gold labels to simulate annotation budgets without new human-subject data collection.

The resulting novelty claim is therefore deliberately operational. Across this representative set of prior work, most LLM-as-annotator studies emphasize label accuracy, human–LLM agreement, cost, or downstream performance, while active-learning work emphasizes selection policies and label efficiency. AnnotateBench instead asks which annotation strategy produces the best downstream model under a fixed budget, where the strategy can be human labeling, active selection, LLM labeling, or a hybrid, and where success is judged by learning curves and cost-efficient strategy-budget choices rather than by one label-agreement score alone.

3 Methodology

3.1 Problem Setup

Let $D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n}$ denote a public training set with gold labels and let D_{test} denote a held-out test set. An annotation strategy s receives access to the unlabeled training inputs and a budget b. The strategy selects a subset $S_{s,b}$ of at most b training examples. The experiment reveals the gold labels for those selected examples, trains a downstream classifier on $S_{s,b}$, and evaluates the classifier on D_{test}.

This setup simulates the outcome of annotation without collecting new human labels. It does not measure human annotation time, disagreement, fatigue, or interface effects. It does, however, isolate the budgeted selection question that arises before annotation begins.

3.2 Annotation Strategies

The benchmark compares four gold-label reveal strategies and supports an LLM annotator strategy:

- **Random sampling**: select examples uniformly at random from the training pool.
- **Uncertainty active learning**: seed the labeled set with class coverage, train an interim model, and select high-entropy examples.
- **Diversity active learning**: select examples to improve coverage in TF-IDF feature space.
- **Hybrid active learning**: combine a diversity-oriented seed with uncertainty-based selection.
- **LLM annotator**: select examples with the random budget policy, ask an OpenAI model for labels and confidence scores, and train the same downstream classifier on the model-generated labels.

Future extensions can add LLM-assisted active selection, simulated noisy annotation, selective human review, and additional active-learning algorithms. These extensions are excluded from the present empirical claims until their cost and quality assumptions are explicit.

3.3 Learning Curves

For each strategy and budget, AnnotateBench records macro F1 and accuracy. The main analysis plots macro F1 against the number of revealed labels. Learning curves make two quantities visible that single-budget reporting hides: the early-budget slope and the point of diminishing returns. These curves support budget recommendations such as the smallest budget that reaches a target macro F1 or the lowest-cost Pareto-efficient configuration.

3.4 Cost and Pareto Frontiers

The benchmark treats annotation cost as a sensitivity parameter rather than a fixed fact. We report low, base, and high human-label cost scenarios calibrated from public requester pricing pages checked on 2026-07-12: Amazon Mechanical Turk requester pricing[1] and Prolific researcher pricing[2]. Human cost is estimated as hourly reward times task time, platform fee, and label redundancy. The resulting per-label costs are \$0.040 for a low-cost MTurk-style single-rater estimate, \$0.095 for a Prolific recommended-pay single-rater estimate, and \$0.428 for a higher-pay Prolific two-rater estimate. The base scenario also includes a small illustrative selection overhead for active-learning strategies: \$0.002 per example for uncertainty sampling, \$0.003 for diversity sampling, and \$0.004 for the hybrid strategy. Random sampling has no selection overhead. For a strategy s, budget b, human-label cost h, and selection overhead o_s, estimated cost is

$$C(s, b) = (h + o_s) \times |S_{s,b}|.$$

A budget-strategy point is Pareto-efficient if no other point has both lower or equal cost and higher or equal macro F1, with at least one strict improvement. Pareto frontiers are intended to help

[1] `https://www.mturk.com/pricing`
[2] `https://www.prolific.com/pricing`

practitioners compare strategy choices under realistic budget constraints. The reported figures use the base scenario; the result CSVs retain all three cost scenarios for sensitivity analysis.

LLM annotation costs are handled separately because they depend on model choice and prompt length. For API-backed annotation runs, estimated API cost is computed from public model prices and recorded token counts:

$$C_{\text{LLM}} = \frac{t_{\text{in}} p_{\text{in}}}{10^6} + \frac{t_{\text{out}} p_{\text{out}}}{10^6}.$$

Managed annotation vendors with quote-based pricing are excluded from the primary quantitative comparison unless a dated quote or invoice is available. Because the LLM annotator cost logs are complete for seeds 1 and 2 but not seed 0, the LLM API costs are reported as an auxiliary token-usage analysis rather than as part of the main human-label Pareto frontiers.

4 Experiment Setup

4.1 Dataset

The benchmark implementation targets ten public text-classification datasets: Financial PhraseBank [9], TREC coarse question classification [8], Banking77 intent classification [1], AG News, SST-2, 20 Newsgroups, Rotten Tomatoes, Yelp Polarity, TweetEval Sentiment, and Emotion. Financial PhraseBank uses a deterministic stratified 80/20 train/test split. TREC and Banking77 use their public train/test splits. The remaining datasets are loaded through HuggingFace and converted into the same text/label interface. Large datasets are capped to keep repeated active-learning runs tractable.

4.2 Model and Metrics

Each condition trains the same downstream text classifier: TF-IDF features with unigram and bigram features followed by logistic regression. Macro F1 is the primary metric because class imbalance can obscure poor minority-class performance. Accuracy is retained as a secondary metric for interpretability.

4.3 Budget Grid and Repetition

The benchmark budget grid is 50, 100, 250, 500, and 1000 labeled examples, capped at the available training set size. This study reports three strategy-selection seeds, 0, 1, and 2. The repeated runs support mean and sample-standard-deviation reporting for the main strategy comparisons; broader repetition and split-sensitivity analyses are left to future benchmark expansions.

4.4 Reproducibility

The repository includes the benchmark runner, data-loading utilities, result-generation scripts, and checked-in aggregate outputs needed to reproduce the tables and figures reported here. The artifact guide documents the exact result files, smoke-test workflow, and claim boundaries for the API-backed LLM annotator extension and row-level reliability case studies.

5 Results

The benchmark uses three strategy-selection seeds, 0, 1, and 2. The combined output contains 1,800 measured rows: ten datasets, four selection strategies, five label budgets, three seeds, and three cost scenarios. Table 2 summarizes the best strategy-budget point for each dataset under the base cost scenario. The ranking varies by dataset: random sampling is strongest on Banking77 and 20 Newsgroups; uncertainty sampling is strongest on Financial PhraseBank and SST-2; and the hybrid strategy is strongest on AG News, Emotion, Rotten Tomatoes, TREC, TweetEval Sentiment, and Yelp Polarity.

Table 2: Best strategy-budget point per dataset under the base cost scenario. Scores are mean macro F1 ± sample standard deviation over three strategy-selection seeds using public gold-label reveal simulation.

Dataset	Best strategy	Macro F1	Budget	Base cost
AG News	Hybrid AL	0.790 ± 0.012	1000	$99.20
Banking77	Random	0.545 ± 0.004	1000	$95.20
Emotion	Hybrid AL	0.233 ± 0.007	1000	$99.20
Financial PhraseBank	Uncertainty AL	0.757 ± 0.016	500	$48.60
Rotten Tomatoes	Hybrid AL	0.683 ± 0.012	1000	$99.20
SST-2	Uncertainty AL	0.706 ± 0.011	1000	$97.20
TREC	Hybrid AL	0.786 ± 0.002	1000	$99.20
TweetEval Sentiment	Hybrid AL	0.363 ± 0.007	1000	$99.20
20 Newsgroups	Random	0.347 ± 0.046	1000	$95.20
Yelp Polarity	Hybrid AL	0.860 ± 0.020	1000	$99.20

The budget recommendation table in `results/budget_recommendations.csv` shows that cost-efficient target attainment is highly dataset-dependent. AG News reaches target macro F1 0.75 with 1000 random labels but does not reach 0.80 in the reported grid, while Yelp Polarity reaches 0.80 with diversity sampling at 1000 labels. Financial PhraseBank reaches 0.75 with uncertainty sampling at 500 labels but does not reach 0.80 in the reported grid. Several harder datasets do not reach 0.50 macro F1 under this simple TF-IDF logistic-regression setup, including Emotion, TweetEval Sentiment, and 20 Newsgroups.

These results support the benchmark framing rather than a universal strategy claim. Active-learning variants help on some datasets, especially Financial PhraseBank, TREC, and Yelp Polarity, but random sampling remains competitive or best on several topic and intent datasets. This matters for annotation planning: strategy choice should be evaluated jointly with dataset, label budget, model family, and cost assumptions rather than selected by default.

LLM annotator extension. We also ran a seed-0/1/2 API-backed LLM annotator strategy for all ten datasets at budgets 50, 100, and 250. This strategy uses random budget selection, asks `gpt-4o-mini` to label the selected training examples with confidence scores, trains the same TF-IDF logistic-regression classifier on the generated labels, and evaluates on the gold test set. Table 3 reports the 250-label point averaged across the three seeds. The results show that LLM label quality and downstream label efficiency are related but not identical. On Financial PhraseBank and Yelp Polarity, LLM labels are highly accurate and downstream performance approaches or matches the best matching gold-label strategy. On high-cardinality or schema-sensitive datasets such as Banking77, Emotion, and 20 Newsgroups, the LLM annotations are substantially less useful for training the downstream classifier at this budget. For the seed-1 and seed-2 annotation logs, recorded token usage totals 2.03M input tokens and 0.21M output tokens, corresponding to an estimated API cost of $0.43 under the published `gpt-4o-mini` prices of $0.15 per million input

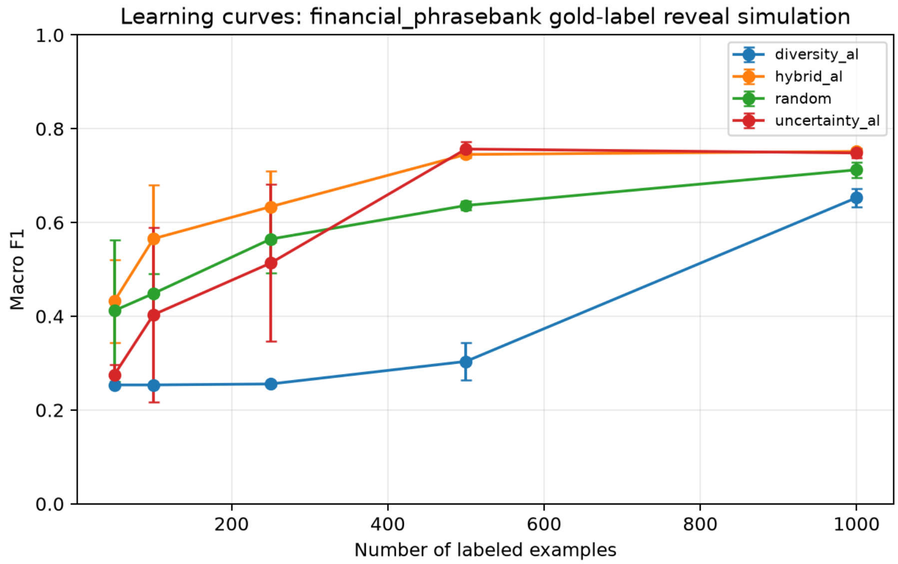

Figure 1: Learning curves for Financial PhraseBank. Uncertainty sampling reaches the best score at a lower cost than the 1000-label alternatives in the reported grid.

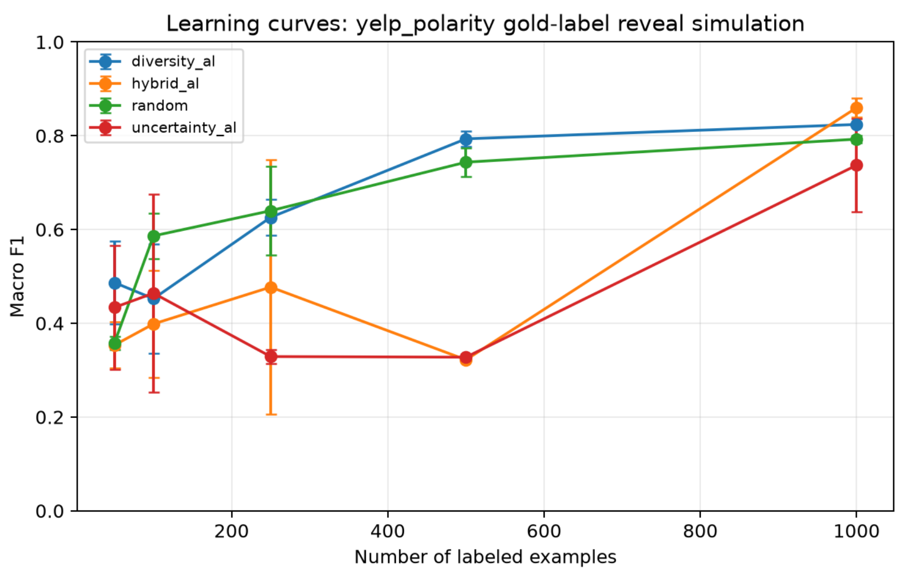

Figure 2: Learning curves for Yelp Polarity. The hybrid and diversity strategies outperform random sampling at higher budgets.

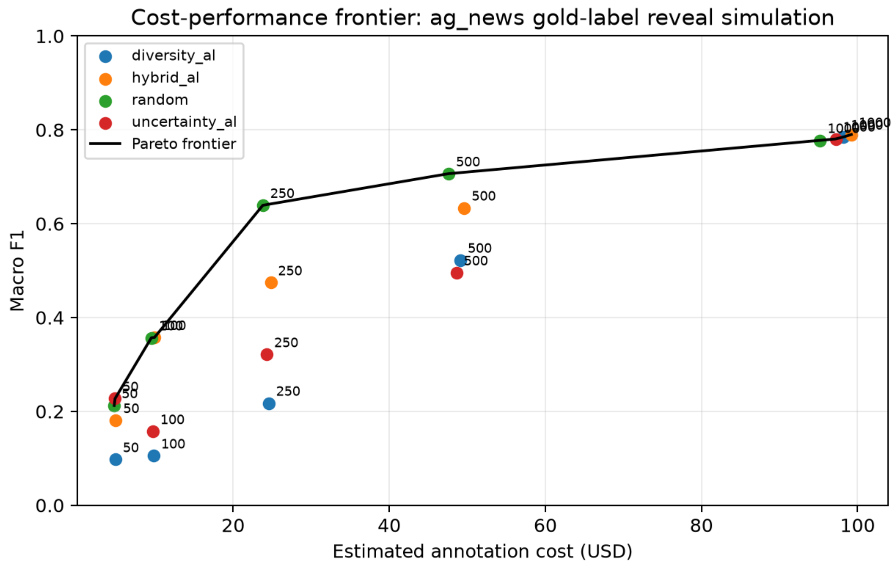

Figure 3: Cost-performance Pareto frontier for AG News under the base cost scenario. The frontier illustrates why strategy choice must be read jointly with budget and cost: random sampling provides cheaper target attainment at intermediate thresholds, while the best overall score in Table 2 comes from hybrid active learning.

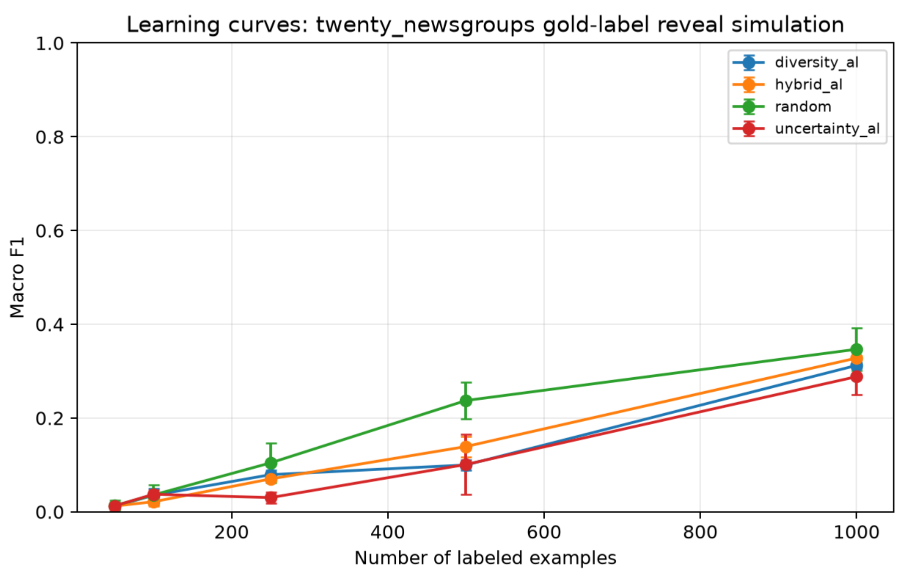

Figure 4: Learning curves for 20 Newsgroups. The simple downstream model remains weak under the reported 1000-label cap, illustrating that annotation strategy cannot compensate for all task/model difficulty.

tokens and \$0.60 per million output tokens. These API costs are reported as an auxiliary analysis because the seed-0 run predates the complete token-logging schema.

Table 3: LLM annotator strategy at budget 250, averaged across seeds 0, 1, and 2, compared with the best gold-label strategy at the same dataset, budget, and seed. Gap is LLM downstream macro F1 minus best gold-label macro F1. LLM label accuracy and LARI are measured on the selected training examples.

Dataset	LLM F1	Best gold F1	Gap	LLM label acc.	LARI
AG News	0.404	0.639	-0.235	0.819	0.781
Banking77	0.084	0.138	-0.054	0.661	0.517
Emotion	0.081	0.158	-0.077	0.584	0.453
Financial PhraseBank	0.589	0.634	-0.045	0.929	0.812
Rotten Tomatoes	0.426	0.588	-0.162	0.908	0.821
SST-2	0.511	0.634	-0.123	0.921	0.838
TREC	0.338	0.528	-0.190	0.691	0.581
TweetEval Sentiment	0.236	0.312	-0.076	0.668	0.591
20 Newsgroups	0.013	0.107	-0.094	0.643	0.548
Yelp Polarity	0.675	0.672	0.003	0.972	0.895

We further ran row-level reliability case studies on Financial PhraseBank, TREC, and TweetEval Sentiment. Each case study annotated 100 held-out test examples with `gpt-4o-mini` at temperatures 0 and 0.7, with three replicates per temperature, for 600 annotations per dataset. Financial PhraseBank is the high-agreement case, with 0.990 accuracy, 0.987 macro F1, ECE 0.161, and LARI 0.828; all six errors are positive examples predicted as neutral. TREC is harder under the same design, with 0.792 accuracy, 0.562 macro F1, ECE 0.084, and LARI 0.514 when schema-external predictions such as `ABBR` are counted as predicted labels. TweetEval Sentiment falls between them, with 0.782 accuracy, 0.766 macro F1, ECE 0.026, and LARI 0.745, and most errors soften negative or positive tweets into the neutral class. Pairwise temperature and replicate comparisons are not significant after Bonferroni correction in all three case studies. These row-level analyses are diagnostic case studies rather than a full LLM reliability benchmark.

Downstream-model robustness. As a robustness check, we also ran the ten-dataset gold-label benchmark with sentence-transformer embeddings followed by logistic regression. This stronger representation improves the best macro F1 on eight datasets, with especially large gains on 20 Newsgroups, Emotion, TweetEval Sentiment, and Banking77, but it underperforms the TF-IDF setup on TREC and is roughly tied on Yelp Polarity. These results reinforce the benchmark framing: annotation strategy choice should be interpreted jointly with the downstream model, and the TF-IDF results should be read as the primary controlled baseline rather than as an upper bound on achievable performance.

6 Limitations and Threats to Validity

This study has several important limitations. First, the benchmark reports three strategy-selection seeds rather than a larger repeated evaluation with tighter confidence intervals. The multi-seed results are more stable than a single-seed run, but they are still not enough to characterize small differences between strategies as definitive.

Second, the downstream model is intentionally simple: TF-IDF features with logistic regression. This creates a controlled comparison across annotation strategies, but stronger neural encoders or instruction-tuned classifiers may change the shape of the learning curves. The sentence-transformer robustness run provides sensitivity evidence rather than a replacement for a broader model-family study.

Third, the cost model is calibrated from public pricing formulas rather than measured annotation invoices, selection runtime, annotation latency, quality-control overhead, or annotator disagreement. The LLM annotator extension includes auxiliary API cost estimates for the seed-1 and seed-2 rows with recorded token usage, but the seed-0 run predates the complete token-logging schema. For that reason, the LLM rows are excluded from the main human-label Pareto frontiers.

Fourth, the gold-label benchmark simulates annotation by revealing existing public labels. This isolates the budgeted selection question, but it does not capture the full process of human annotation, including ambiguous instructions, inconsistent annotators, interface effects, fatigue, or adjudication. The LLM annotator extension similarly evaluates one model and one prompt family rather than a full reliability study across models, prompts, domains, and human review policies.

Finally, the measured benchmark is classification-only. NER, QA, summarization, relation extraction, and other structured or generative tasks require task-specific annotation units, metrics, and quality-control workflows. Results from the text-classification suite should not be generalized to those task families without separate runners and evidence.

7 Future Work

Future work should strengthen AnnotateBench in two main directions. First, measured annotation costs and reliability evidence would make the benchmark more useful for deployment decisions. Future evaluations could incorporate observed human-labeling costs, API-backed LLM annotation costs, annotation latency, and quality-control overhead, while also studying LLM confidence calibration and common annotation failure modes.

Second, the benchmark could be expanded beyond the current classification setting. Comparisons with stronger downstream learners and alternative representations would test whether the observed strategy rankings hold under more expressive models. Longer-term extensions could adapt the framework to structured and generative tasks, where annotation units, evaluation metrics, and quality-control workflows differ from text classification.

8 Conclusion

AnnotateBench frames text-classification annotation strategy selection as an empirical budget-allocation problem. Instead of asking only which model achieves the highest score, it asks how performance improves as labels are revealed and which strategy-budget combinations are cost-efficient. This JDSE 2026 submission establishes the methodology, novelty positioning, and ten-dataset classification evidence for a reusable benchmark of annotation strategy choice.

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